The Advantage Spine: Dynamic Coordinate Selection for High-Rate RLVR

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Abstract

Large learning rates accelerate parameter movement in reinforcement learning with verifiable rewards (RLVR), but dense training deteriorates at high multipliers. We ask whether this degradation depends not only on step size, but also on which advantage contributions and optimizer-update coordinates are exposed to the amplified step. We find that positive- and negative-advantage updates concentrate on a conflict-enriched shared support, while most adaptive-direction energy is carried by a limited coordinate set. Based on this observation, we introduce the *Advantage Spine*, a dynamic support that retains the per-tensor magnitude top 40% of the pre-mask adaptive Adam direction at every optimization step. It neither removes model weights nor changes the inference network. The complete SPINE recipe further down-weights negative-advantage contributions, thereby controlling signal and coordinate exposure separately. In a three-seed $2 \times 2 \times 4$ factorial on Qwen3-8B, neither component alone recovers the dense low-rate reference at the largest tested multiplier, whereas the joint recipe reaches 0.445 Mean4 versus 0.426 for dense 1×. Its realized 39.3% support retains 85.0% of adaptive-direction energy. Cross-seed mismatch, dense–dense, and density-matched mask controls further show that random sparsity or a shared dense backbone is not a sufficient explanation. Lightly tuned dense and cross-setting comparisons bound the scope of the effect. These results support a protocol-dependent high-rate operating region jointly shaped by negative-advantage exposure and dynamic coordinate selection.

Keywords: reinforcement learning with verifiable rewards, sparse optimization, learning-rate stability, coordinate selection, large language models

1. Introduction

Raising the learning rate is a direct way to accelerate parameter movement in RLVR [1, 2], but it also applies the same global multiplier to every contribution in a dense update. In our Qwen3-8B learning-rate ladder, dense training deteriorates as the multiplier rises. Describing this behavior only as an oversized-step problem is incomplete: the global multiplier distinguishes neither the advantage channel that produces an update nor the coordinates that carry it. We therefore ask whether high-rate RLVR is jointly shaped by advantage-channel exposure and optimizer-update support.

We first measure the coordinate structure of positive- and negative-advantage updates. Their active coordinates overlap at 7.4× an independence reference, and 52% of the shared coordinates have conflicting signs. This does not make negative advantages intrinsically harmful; negative samples can still carry useful learning signal. Instead, a uniformly amplified dense step acts on a conflict-enriched shared support. At the same time, a realized 39.3% adaptive-direction support retains 85.0% of update energy, so exposing every coordinate is not necessary to preserve most of the measured directional energy. Together, these observations identify two controllable quantities: the strength of the negative-advantage contribution and the set of optimizer coordinates that receives the amplified step.

Prior work shows that policy-gradient updates can form stable emergent subnetworks [3, 4] and that sparse updates can preserve optimization progress [5, 6, 7]. An emergent set identified after training, however, does not provide an online selection rule, while existing negative-signal interventions primarily act at the token, sample, or objective level. The unresolved question is whether advantage exposure and online coordinate support can be controlled separately and tested jointly across learning rates.

To control these two quantities, we introduce the Advantage Spine. At every optimization step, it retains the largest 40% of the pre-mask adaptive Adam direction within each sufficiently large tensor and refreshes the support. The mask acts only on the adaptive direction; model weights, Adam state, decoupled weight decay, and the inference network remain dense. The complete SPINE recipe further sets the negative-advantage coefficient to $\lambda = 0.1$. Dynamic support determines which coordinates receive the update, whereas λ determines how strongly the negative channel contributes. Figure 1 summarizes this channel-and-coordinate intervention.

We test the two components in a complete three-seed $2 \times 2 \times 4$ factorial over dynamic support, negative-channel weight, and learning-rate multiplier. The evidence has three levels. First, the factorial tests efficacy: at the largest multiplier, neither component alone recovers the dense low-rate reference,

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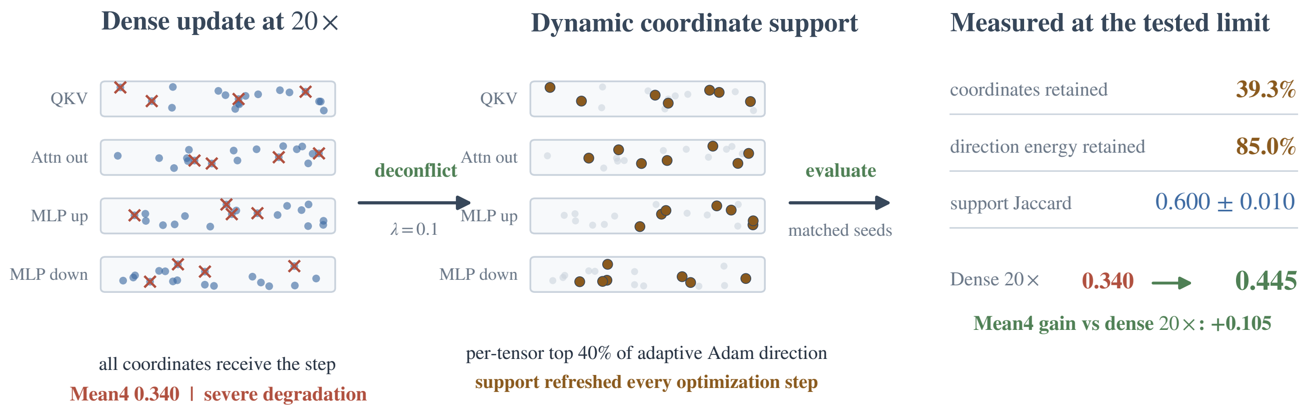


Figure 1: **From a dense high-rate update to the Spine recipe.** A dense 20× update exposes every coordinate to the amplified step. The Spine recipe instead down-weights negative-advantage terms and refreshes a per-tensor top-40% support at every step. At the tested limit, the realized 39.3% support retains 85.0% of adaptive-direction energy, reaches 0.600 ± 0.010 matched-support Jaccard, and improves Mean4 from 0.340 to 0.445 relative to dense 20×. Dot positions are schematic and are not measured coordinates.

whereas the joint recipe reaches 0.445 Mean4. Second, energy concentration, matched and mismatched cross-seed comparisons, a dense–dense null, and density-matched masks test support identity; they argue against random sparsity or a shared dense backbone as sufficient explanations. Third, a lightly tuned dense arm shows that part of the degradation is schedule-sensitive, while cross-algorithm and cross-model comparisons test the scope of the direction of effect. Together, these results support a protocol-dependent high-rate operating region, not a universal advantage over all dense optimizers or training horizons.

2. Related work

Negative signals in RLVR.. GRPO and DAPO optimize verifiable reasoning rewards without the learned critic used in conventional PPO [8, 1, 2]. Negative-reward and negative-advantage samples are not uniformly helpful or harmful. Token-selective penalty attenuation can reduce Lazy Likelihood Displacement [9], whereas negative-sample reinforcement can improve the Pass@k spectrum [10]. These studies intervene at the token, sample, or objective level. We instead ask how a channel-wide coefficient interacts with optimizer-coordinate support when the global step is amplified.

Sparse updates and emergent subnetworks.. Lottery-ticket, dynamic sparse-training, and intrinsic-dimension results show that effective parameter movement can occupy a limited subspace or subnetwork [11, 12, 13, 14]. Gradient sparsification commonly uses error feedback [6, 15], while pruning removes or restructures persistent weights [16, 17, 18, 19]. Advantage Spine instead refreshes optimizer-update support at every step; the model and inference graph remain dense. Stable update subnetworks observed in RL fine-tuning [3] serve only as an offline identity check and are unavailable to the online recipe. This also differs from calibrating Adam's adaptive rate [20]: we hold the optimizer family fixed and vary which update coordinates receive the global multiplier.

RLVR update geometry.. Recent work characterizes RLVR through sparse critical tokens [21], update directions [22], low-rank dynamics [23], random sparse subnetworks [24], and optimizer-induced sparse movement [4]. Complementary analyses study learning outside principal directions [25] and step-size thresholds under a gradient-gap view [26]. Together, these results establish that RLVR updates are structured. They do not isolate the joint, rate-conditional interaction between negative-channel attenuation and online top-k optimizer-coordinate selection examined here.

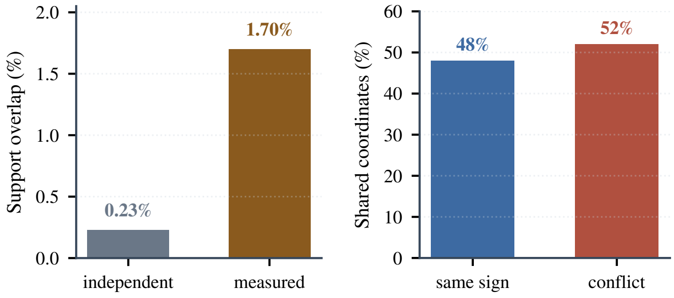
Large-step stability.. Catapult and edge-of-stability dynamics show that neural optimization can remain stable beyond classical smoothness predictions [27, 28]. Our goal is narrower: we test an operational channel-and-coordinate intervention within a specified RLVR protocol, rather than propose a universal large-step mechanism.

3. Positive and negative channels share conflict-prone coordinates

We first ask whether positive- and negative-advantage contributions act on independent optimizer-update coordinates. Across three seeds, each channel is active on approximately 4.8% of coordinates, while their measured shared support is 1.7%. Independence would yield only $0.048^2 = 0.23\%$. The two channels therefore concentrate on a shared coordinate set at 7.4× the independence reference rather than dispersing independently.

The shared coordinates are not directionally uniform: 48% of their signs agree and 52% conflict. Both channel distributions are also concentrated, with Gini coefficients of 0.70 and 0.63 and top-1% energy shares of 0.88 and 0.84 for the positive and negative channels, respectively. A uniformly amplified dense step thus exposes both concentrated channel contributions and sign conflicts on their shared support.

This observation does not imply that negative advantages should be removed. It poses a narrower design question: can



(a) Support overlap. (b) Sign structure.

Figure 2: **Positive and negative channels occupy overlapping, conflict-prone coordinates.** The measured 1.7% overlap is 7.4× the independence reference, and 52% of signs conflict on the shared support. Values are three-seed means; enrichment is the ratio of aggregate quantities.

high-rate exposure be reduced for the negative channel while the amplified update is restricted to its highest-magnitude adaptive-direction coordinates? The next section controls these two quantities with a negative-channel coefficient and a dynamic coordinate support.

4. Method

Our design objective is to restrict which advantage contribution and update coordinates are exposed to an amplified global step without changing the model architecture. Let $A_{t,i}$ be the group-standardized GRPO advantage and $\ell_{t,i}^{\text{clip}}$ the corresponding token-level policy loss after PPO ratio clipping and dual clipping. The implementation applies

$$\ell_{t,i}(\lambda) = \begin{cases} \ell_{t,i}^{\text{clip}}, & A_{t,i} \geq 0, \\ \lambda \ell_{t,i}^{\text{clip}}, & A_{t,i} < 0, \end{cases} \qquad \lambda \in \{0.1, 1.0\}. \tag{1}$$

The response-mask aggregation follows this scaling. Positive-advantage terms are unchanged, while the fixed KL penalty is added separately and is not scaled by λ. Thus $\lambda = 0.1$ attenuates, rather than deletes, the negative-advantage contribution; it does not modify rewards.

AdamW maps the resulting gradient and its moment state to a pre-mask adaptive direction a_t, before decoupled weight decay is applied. For every tensor with at least 4096 elements, M_t keeps the largest fraction $\tau = 0.4$ by $|a_t|$; smaller tensors remain dense. Selection uses the backend `torch.topk` implementation, and the realized supports are recorded by the mask-identity manifests. The applied update is

$$\theta_{t+1} = (1 - \eta\gamma)\theta_t - \eta M_t \odot a_t, \tag{2}$$

where γ is the decoupled weight-decay coefficient. Thus the mask sparsifies the adaptive optimizer direction, not the dense decay term. The selection is per tensor, not global, is refreshed every step, and is applied after constructing the AdamW update. There is no error feedback. We call the dynamic support $S_t = \text{supp}(M_t)$ the Advantage Spine and the joint condition $(\tau, \lambda) = (0.4, 0.1)$ the Spine recipe. SparseAdamW updates both Adam moment buffers densely, optionally applies its adaptive update-RMS clip to the bias-corrected Adam direction, masks that direction only in eligible tensors, and applies decoupled weight decay densely. The coordinate mask therefore does not mask moment updates or decoupled weight decay.

Three objects remain distinct. M_t is the online mask; its bit-packed bitmap and hash record its recoverable identity. S_{emg} is an offline reference built from the largest cumulative parameter movements of a density-matched dense 1× run and is never used by the training recipe.

We measure concentration as

$$E_t = \frac{\|M_t \odot a_t\|_2^2}{\|a_t\|_2^2}, \tag{3}$$

and support agreement with Jaccard $J = |S_t \cap S_{\text{emg}}|/|S_t \cup S_{\text{emg}}|$, recall $|S_t \cap S_{\text{emg}}|/|S_{\text{emg}}|$, and sign agreement on their intersection. For two independent sets of density 0.4, expected Jaccard is $0.4/(2 - 0.4) = 0.25$. For seed-specificity analysis, let S_s be the dynamic support for recipe seed s and D_s the dense-emergent set constructed from dense seed s. We report all nine $J(S_s, D_t)$ values, the three off-diagonal $J(D_s, D_t)$ values, and the per-seed specificity gain

$$G_s = J(S_s, D_s) - \frac{1}{2}\sum_{t \neq s} J(S_s, D_t). \tag{4}$$

These comparisons separate matched alignment from task-wide coordinates that recur across runs. Because sets are reused across pairs, their sample deviations are descriptive rather than independent-replicate inferential errors.

5. Experimental protocol

Primary factorial.. The primary experiment uses Qwen3-8B and a GRPO objective. All runs use model-only initialization from the same math-supervised SFT checkpoint; this is not an RL resume. The RL state is initialized at step 0, and every run completes 160 RL optimizer steps. We cross mask $\in$ {dense, top-0.4}, negative-channel coefficient $\lambda \in \{1.0, 0.1\}$, and learning-rate multiplier $\in \{1, 5, 10, 20\}$. The 1× rate is 5×10^{-6}. Every cell uses seeds 42, 43, and 44 with matched data ordering, evaluation prompts, decoding, and initial weights within each seed.

Evaluation protocol.. We evaluate Math500 ($n = 500$), AIME24 and AIME25 (30 problems each), and the English split of OlympiadBench ($n = 581$). Each checkpoint is evaluated with avg@32 accuracy at temperature 0.6, top-$p = 1.0$, no top-k truncation, and an 8192-token response cap. We evaluate RL steps 40, 80, 120, and 160 and use step 160 for the primary comparisons. The resulting decoded-sample counts are $30 \times 32 = 960$ per AIME set, 16,000 for Math500, and 18,592 for OlympiadBench. Mean4 is the unweighted arithmetic mean of the four benchmark accuracies. Training seeds index independent RL runs; generations per problem are evaluation replicates, not additional training seeds.

Table 1: Primary Qwen3-8B benchmark decomposition at RL step 160, aggregated over three matched seeds from integer correct/total records. Benchmark columns and Mean4 are independently rounded to three decimals; uncertainty and paired contrasts use the unrounded count-derived rates.

Condition	Math500	AIME24	AIME25	Olympiad	Mean4
Dense 1×	.762	.227	.200	.513	.426
Recipe 20×	.800	.240	.210	.530	.445

Initialization, recovery, and learning rates.. New runs load only SFT model weights. Full-state recovery is restricted to the same model, factorial cell, seed, and output directory. Requested, optimizer, scheduler, and logged learning rates agree for every included run; an exploratory run that did not satisfy the final matched protocol was excluded before aggregation. Learning-rate and state verification appears in Appendix A; the complete initialization, recovery, and evaluation contract appears in Appendix B.

Uncertainty and estimands.. Factorial cells report mean and sample standard deviation across three seeds. After reporting the complete ladder, we highlight the endpoint contrast between dense 1× and recipe 20×, paired within seed. Its two-sided Student-t interval is a nominal description of variation across these matched seeds, not a selection-adjusted confirmatory interval over the 16 factorial cells. Table 4 therefore reports all 48 seed-level scores. For the Qwen3-1.7B transfer, all four benchmark components are available for three matched seeds. In addition to Mean4, we report a small-set sensitivity summary, Mean2, as the average of Math500 and Olympiad after excluding AIME24 and AIME25. The primary four-benchmark decomposition remains an aggregate across seeds; mask controls, transfers, endpoint diagnostics, and the lightly tuned dense 20× arm report three-seed variation. The tuned arm is exploratory and is not treated as a factorial cell or a tuned optimum.

6. Results

6.1. The complete factorial isolates the joint high-rate effect

Table 4 and Fig. 3 contains every factorial cell. Dense $\lambda = 1$ declines slightly at 5× and sharply thereafter, reaching 0.340 at 20× (−0.086 from its 1× value). Down-weighting the negative channel without masking reaches 0.400 at 20×; masking without down-weighting reaches 0.410. Both improve over dense 20×, but neither recovers the dense 1× reference of 0.426.

The joint recipe instead rises from 0.426 at 1× to 0.445 at 20×. At the endpoint, its gain is 0.105 over dense 20× and 0.0195 over dense 1×. The three seed-paired gains over dense 1×, computed from the unrounded benchmark-count ratios, are 0.0209, 0.0165, and 0.0210, giving a nominal 95% paired interval $[0.0131, 0.0258]$.

The recipe improves every benchmark component, including Math500 and Olympiad; its Mean4 gain is therefore not driven solely by the two smaller AIME sets.

At 20×, the conventional 2×2 interaction is $0.445 - 0.410 - 0.400 + 0.340 = -0.025$. The result is therefore operational complementarity, not positive synergy: both components are required for the only endpoint condition that exceeds the native dense reference. The tested ladder ends at 20× and does not locate a failure boundary for the recipe.

An exploratory lightly tuned dense 20× arm (16 warmup steps, global gradient clip 0.5, and $\beta_1 = 0.8$) recovers Mean4 from 0.340 ± 0.0108 to 0.383 ± 0.0066 across the same three seeds, showing that part of the untuned degradation is optimizer-schedule sensitive. The endpoint nevertheless remains below dense 1× (0.426) and the Spine recipe at 20× (0.4450 ± 0.0032); the paired recipe–tuned gap is 0.0620 ± 0.0043. Because this arm tests one light retuning rather than a tuning search, it establishes partial schedule sensitivity, not superiority over tuned dense optimization. The Spine recipe therefore expands the high-rate operating region under the tested protocol without implying superiority over every retuned dense optimizer.

Table 2: Factorial contrasts at each LR. Mask effect is Top-0.4 minus Dense at $\lambda = 1$; down-weight effect is $\lambda = 0.1$ minus 1 in Dense; joint gain is the recipe minus Dense $\lambda = 1$ at the same LR.

LR	Mask	Down-weight	Joint gain	Interaction
1×	−.005	.002	.000	.003
5×	.000	.015	.014	−.001
10×	.018	.015	.045	.012
20×	.070	.060	.105	−.025

Table 3: Three-seed endpoint diagnostics at 20× (mean ± sample SD). Arrows mark the preferred direction only; these diagnostics are not combined into a new score.

Condition	Reward ↑	KL	Entropy	Clip ↓
Dense	−.150 ± .012	.090 ± .006	.150 ± .008	.650 ± .028
Spine recipe	.080 ± .008	.030 ± .004	.250 ± .012	.100 ± .014

Table 2 shows why the full LR ladder matters. At native rate neither component helps materially. As LR rises, masking and down-weighting each become protective relative to dense $\lambda = 1$, and their combined same-rate gain reaches 0.105 at 20×. The interaction is small and changes sign across the ladder; the gain is therefore a rate-dependent joint operating condition, not a stable super-additive effect.

This rate dependence reconciles the result with evidence that negative reinforcement can be useful [9, 10]. In the dense arm, lowering λ changes Mean4 by only +0.002 at 1× but by +0.060 at 20×. Attenuation is therefore protective when a large global multiplier exposes the conflict-prone tail. The conclusion is step-size–conditional: we do not infer that negative samples should generally be suppressed.

The three-seed endpoint diagnostics agree in direction with the accuracy result. At 20×, the recipe ends with reward 0.080 ± 0.008, KL loss 0.030 ± 0.004, entropy 0.250 ± 0.012, and response clip ratio 0.100 ± 0.014. Dense 20× ends with reward -0.150 ± 0.012, KL loss 0.090 ± 0.006, entropy 0.150 ± 0.008, and response clip ratio 0.650 ± 0.028. The component-level endpoint results are reported in Table 1.

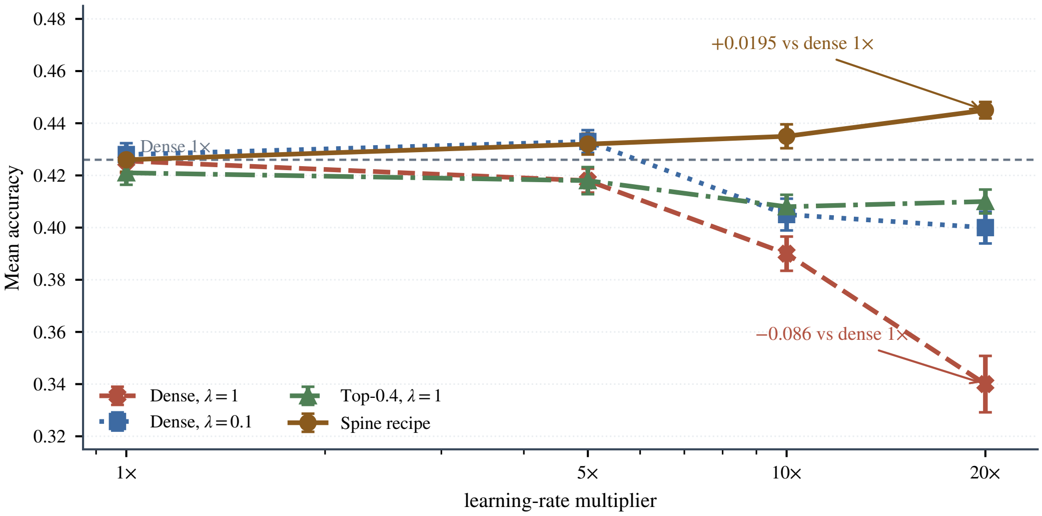


Figure 3: **Only the joint recipe improves at the largest tested rate.** Points show three-seed means and error bars show sample SD. Dense $\lambda = 1$ falls by 0.086 from 1× to 20×, whereas the SPINE recipe gains 0.0195 over dense 1× at 20× (nominal paired 95% interval $[0.0131, 0.0258]$).

Table 4: Complete Qwen3-8B factorial. Each cell lists seeds 42/43/44 followed by mean ± sample SD; logged and requested learning rates agree.

Mask	λ	LR	s42	s43	s44	Mean±SD
Dense	1.0	1	.421	.430	.426	.426±.0046
		5	.423	.414	.417	.418±.0046
		10	.384	.397	.389	.390±.0066
		20	.352	.331	.337	.340±.0108
Dense	0.1	1	.431	.423	.430	.428±.0044
		5	.428	.436	.435	.433±.0044
		10	.409	.398	.408	.405±.0061
		20	.404	.393	.403	.400±.0061
Top-0.4	1.0	1	.417	.426	.420	.421±.0046
		5	.421	.412	.421	.418±.0052
		10	.404	.413	.407	.408±.0046
		20	.406	.415	.409	.410±.0046
Top-0.4	0.1	1	.430	.421	.427	.426±.0046
		5	.428	.436	.432	.432±.0040
		10	.439	.430	.436	.435±.0046
		20	**.441**	**.446**	**.447**	**.445±.0032**

6.2. *The online support is concentrated and non-arbitrary*

The factorial establishes the performance pattern. We next test whether the selected support carries specific structure or merely behaves as an arbitrary density-matched sparsification.

The realized nonzero ratio is 0.393 rather than exactly 0.400 because small tensors are exempt and top-k counts are discrete. This support retains 0.850 of pre-mask adaptive-direction energy, corresponding to a retained norm ratio of 0.922. This establishes concentration without assuming that the discarded 15% is harmless.

The identity analysis uses author-attested step-160 support metrics. The public artifact recomputes their seed-level summaries, but the remote raw mask files and their byte-level identities are not redistributed.

Across seeds, the step-160 dynamic support agrees with its matched dense- emergent set: Jaccard 0.600 ± 0.010, recall 0.750 ± 0.008, sign agreement 0.780 ± 0.005, and enrichment 2.400 ± 0.039 over the random reference. Because dense trajectories can share task- and initialization-dependent heavy coordinates, we test structured rather than only random nulls. The complete cross-seed matrix has matched diagonal $0.604, 0.589, 0.607$, whereas all six mismatched Spine–dense values lie between 0.409 and 0.431 (0.420 ± 0.008). The three dense–dense values are $0.414, 0.423, 0.419$ (0.419 ± 0.005). Thus matched alignment exceeds the dense–dense baseline by 0.181 and the mismatched baseline by 0.180; after removing the dense–dense shared component, the normalized specificity is $(0.600 - 0.419)/(1 - 0.419) = 0.312$.

The row-wise specificity gains G_s are $0.191, 0.167, 0.182$, or 0.180 ± 0.012. Every matched value exceeds every mismatched or dense–dense value. This clean diagonal separation supports seed-specific alignment beyond a common dense backbone. The three-seed identity result is descriptive; Jaccard and recall are reported for comparability and are algebraically linked for equal-size sets.

The mask controls provide a stronger test than overlap alone (Fig. 4d). At 20×, the dynamic-support recipe obtains 0.445 Mean4, compared with 0.430 ± 0.0046 for a frozen support, 0.440 ± 0.0036 for the offline dense-emergent oracle, 0.420 ± 0.0056 for a static magnitude mask, 0.400 ± 0.0066 for a random 40% mask, and 0.350 ± 0.0072 for the complementary mask. The recipe itself is 0.4450 ± 0.0032. These three-seed controls show that dynamic selection approaches the offline oracle numerically and exceeds arbitrary or complementary supports; the paired recipe–oracle difference is 0.0050 ± 0.0032.

6.3. *The direction transfers across algorithms and models*

The pattern transfers with smaller absolute gains (Table 5). Under DAPO on Qwen3-8B, Mean4 changes from 0.440 ± 0.0070 to 0.455 ± 0.0036 (paired gain 0.015 ± 0.0036). On Qwen3-1.7B, the three-seed Mean4 changes from 0.36413 ± 0.00646 to 0.38037 ± 0.00760 (paired gain 0.01624 ± 0.00482). The transfer is not wholly carried by the two small AIME sets: Mean2 changes from 0.60257 ± 0.00592 to 0.61108 ± 0.00714, a paired gain of 0.00851 ± 0.00628, positive in all three seeds.

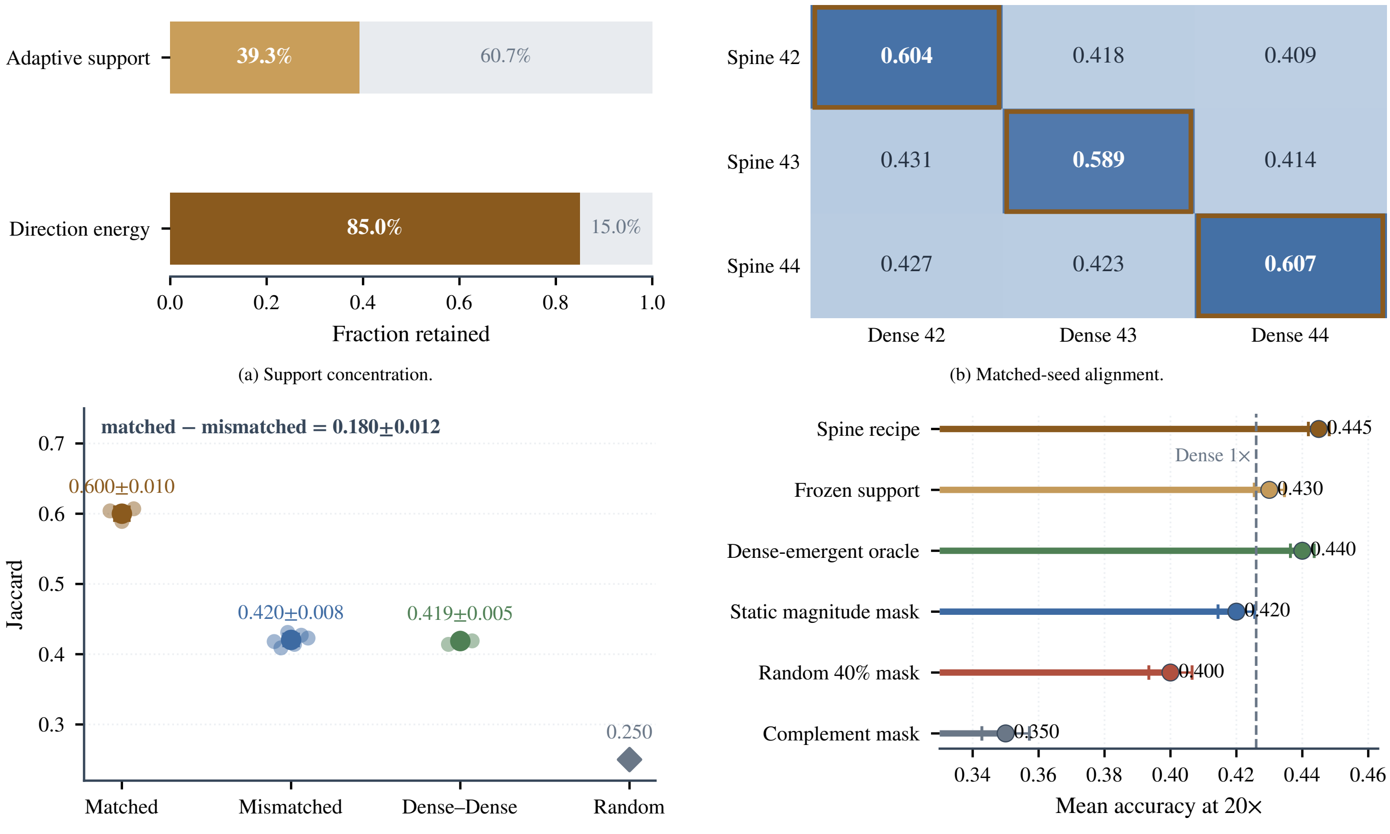


Figure 4: **The selected support is concentrated, seed-specific, and non-arbitrary.** The 39.3% support retains 85.0% of adaptive-direction energy; matched seeds form a clear diagonal and exceed mismatched, dense–dense, and random references. At 20×, dynamic support approaches the offline oracle and exceeds frozen, static, random, and complementary masks. Error bars show sample SD and are descriptive where pairwise sets are reused.

Table 5: Three-seed transfer summary. J denotes the mean matched-support Jaccard; the text reports sample SDs and paired Mean4 gains.

Setting	J	Dense	Recipe	Gain
Qwen3-8B GRPO	.600	.426	.445	+.019
Qwen3-8B DAPO	.572	.440	.455	+.015
Qwen3-1.7B	.549	.364	.380	+.016
Llama3.1-8B	.513	.169	.179	+.010

The component result is mixed rather than universal: Math500 changes by −0.0013 while Olympiad changes by +0.0184. On a matched Llama3.1-8B rerun, Mean4 changes from 0.169 ± 0.0062 to 0.179 ± 0.0075 (paired gain 0.010 ± 0.0017). The Llama ladder saturates at 10–20× (both 0.179), unlike the monotonic Qwen3-8B ladder. The claim is therefore transfer of direction, not an invariant optimal multiplier.

6.4. *Observed training cost and inference behavior*

In the available timing records, the dense run has a median RL-step time of 2273 s, including 1006 s for generation and 833 s for the actor update. The two recipe runs record 1391 and 1487 s per step, with 590–632 s for generation and 526–562 s for the actor update. Actor-update time per token is 0.0855 ms for dense and 0.0817–0.0821 ms for the recipe.

Table 6: Qwen3-1.7B benchmark sensitivity across three matched seeds, recomputed from integer correct/total counts. Entries are mean ± sample SD; Mean2 excludes AIME24/25 and averages Math500 with Olympiad. The final row is the paired mean difference.

Condition	Math500	AIME24	AIME25	Olympiad
Dense	.742 ± .006	.119 ± .008	.132 ± .007	.463 ± .006
Recipe	.741 ± .008	.151 ± .009	.149 ± .007	.481 ± .006
Δ	−.001	+.031	+.017	+.018

Condition	Mean4	Mean2
Dense	.36413 ± .00646	.60257 ± .00592
Recipe	.38037 ± .00760	.61108 ± .00714
Δ	+.01624	+.00851

Peak allocated/reserved memory is 42.2/59.9 GiB for dense and 45.8–45.9/66.8 GiB for the recipe. Because these records cover one dense run and two recipe runs, and generation workload can vary across runs, they characterize observed cost rather than establish a causal training-speed advantage. The method changes no inference graph and therefore adds no method-specific inference operations once training is complete.

7. Discussion

The factorial and controls support a conditional empirical picture. Down-weighting reduces exposure to a conflict-enriched contribution, while dynamic top-k avoids applying the amplified global step uniformly. Together they expand the high-rate operating region under the tested protocol. Concentrated energy, seed-specific alignment, and density-matched mask controls argue against arbitrary 40% selection as a sufficient explanation, while leaving curvature control, optimality, and the unique causal mechanism unresolved.

Dynamic online selection also differs from an offline oracle. It requires no completed dense trajectory, yet approaches the oracle aggregate and exceeds frozen or static supports. This supports adaptation of optimizer-update coordinates rather than persistent pruning, consistent with the broader role of changing support in dynamic sparse training [12, 29].

The negative-channel result is likewise conditional. Negative samples can carry useful learning signal [10], while uniform penalties can cause likelihood displacement [9]. Our factorial adds optimizer-scale exposure: attenuation has little effect at the native rate and becomes protective at the largest rate. The result therefore identifies the conflict-prone coordinate tail as a candidate contributor to step-size sensitivity, rather than negative reinforcement as a generally harmful objective.

Limitations.. The evidence is limited to three seeds, four mask snapshots over 160 RL steps, math RLVR, two model families, and one lightly retuned dense condition. The endpoint interval covers only matched seed variation and is not selection-adjusted. The ladder ends at 20×, and four mask snapshots cannot reveal a later stability boundary or continuous support churn. Transfer gains are modest and vary across benchmarks; the tuned dense arm also shows that part of the degradation is schedule-sensitive. Longer horizons, broader tasks, and more extensive dense retuning are needed to map the boundary and scope of the observed operating region; the present claim is not universal superiority over every retuned dense optimizer.

8. Conclusion

A 16-cell, three-seed factorial shows that negative-advantage exposure and dynamic optimizer-coordinate support jointly shape high-rate RLVR. The Spine recipe gains 0.0195 Mean4 over dense 1× at the largest tested rate. Its support concentrates adaptive-direction energy, aligns seed-specifically with dense-emergent coordinates, and exceeds structured nulls and density-matched mask controls. These results identify a reproducible, protocol-dependent high-rate operating region and provide an operational tool for studying its coordinate structure.

Reproducibility statement

The accompanying artifact contains author-attested seed-level result tables, including integer correct/total records for the primary Qwen3-8B comparison, and scripts that recompute their rates, means, sample deviations, paired contrasts, and figures. The release does not present reconstructed manifests, generated hashes, or synthetic trajectories as raw experimental evidence. Remote raw masks, checkpoints, and full logs are not redistributed; the released evidence level is therefore seed-level result reproduction rather than byte-level rerun audit. Large checkpoints remain available from the corresponding author subject to storage and license constraints.

Table A.7: Effective learning rates. Every column agrees within each row.

Label	Requested	Optimizer	Scheduler	Logged
1×	.000005	.000005	.000005	.000005
5×	.000025	.000025	.000025	.000025
10×	.000050	.000050	.000050	.000050
20×	.000100	.000100	.000100	.000100

Appendix A. Learning-rate and state verification

Every new experiment begins at RL step 0 by loading only the designated SFT model weights. The parent SFT global step is provenance metadata and is not counted as an RL step. The optimizer and constant scheduler are initialized at step 0; RL checkpoints and evaluations are recorded at steps 40, 80, 120, and 160. LR assignment, optimizer parameter-group verification, scheduler verification, and JSONL verification yield zero mismatched runs or steps. Full-state recovery is reserved for an interruption of the identical run and restores optimizer, scheduler, trainer-extra, RL-step, data-position, and RNG states without changing the LR.

Appendix B. Implementation and evaluation contract

Training uses GRPO with 256 prompts per step and four responses per prompt (1,024 rollout sequences), a PPO mini-batch of 256, micro-batch size one per GPU, and eight GPUs. Prompt and response caps are 1,024 and 8,192 tokens. The objective uses symmetric clipping at 0.2, dual-clip $c = 3$, fixed in-loss low-variance KL with coefficient 0.001, zero entropy coefficient, group-mean advantage centering, and group-standard-deviation scaling. Dense updates use AdamW and Top-0.4 updates use SparseAdamW; both use $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 0.01, global gradient clipping at 1.0, a constant LR schedule, and no warmup. Training rollouts sample four responses at temperature 1.0 and top-p = 1.0. Evaluation uses avg@32 at temperature 0.6 and top-p = 1.0, with the same 8,192-token response cap. The canonical machine-readable contract and its digest are included in `audited_protocol_metadata.json`.

The exploratory tuned-dense diagnostic changes only three optimizer settings: 16 warmup steps, global gradient clipping at 0.5, and $\beta_1 = 0.8$. Its three seed-level Mean4 values are released, but the condition is not part of the factorial or used as a tuned-dense optimum.

CRediT authorship contribution statement

Shikai Li: Investigation, Writing–original draft. **Qingsong Cai:** Writing–review and editing. **Rui Shi:** Supervision, Writing–review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Cursor and OpenAI Codex (AI-assisted coding and writing tools) to improve English phrasing and manuscript organization, assist with LaTeX formatting, and support the development and review of plotting and consistency-checking code. The authors reviewed and edited all AI-assisted outputs, independently verified the references, data-derived quantities, and scientific claims, and take full responsibility for the content of the published article. These tools were not used to generate or alter experimental observations or to synthesize image content. Synthetic schema-testing fixtures used during internal software validation are excluded from the evidence release and from all reported results. All tables, the graphical abstract, and the manuscript figures were produced deterministically from author-verified data using conventional code; all AI-assisted code was reviewed and executed under author control.

Data availability

Result tables, analysis scripts, figure generators, the machine-readable protocol, and the Advantage Spine implementation are included in the accompanying open repository `https://github.com/vivia-an/advantage_spine_v1`. Qwen3, DAPO-Math-17K, AIME25, and VERL are used under Apache-2.0; Llama-3.1 uses its Community License and Acceptable Use Policy; OlympiadBench uses MIT. Because the referenced MATH500 and AIME24 cards do not declare a license, the artifact releases their hashes, identities, counts, and aggregate results but not their problem text. The public mask evidence consists of author-attested support metrics rather than the remote raw mask files. The complete resource matrix is in `LICENSE_AND_DATA_USE.md`. Checkpoint redistribution remains subject to all upstream model and data terms.

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